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Agent-Based Model Report
Title: Underwater Trash-Collecting Robots
Group Name: Parallel

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1. Introduction

For our project, we chose SeaClear, a European project that uses a team of robots to find and collect litter from the seabed. The full system has a surface boat (SeaCat), a drone (SeaHawk), an observation ROV (miniTortuga), a collection ROV (Tortuga) and a collection basket. In our model, the observation ROV (ObsROV) searches the area and maps the litter, while the collection ROV (ColROV) picks up the mapped litter and brings it back to the basket.

2. How the Model Works

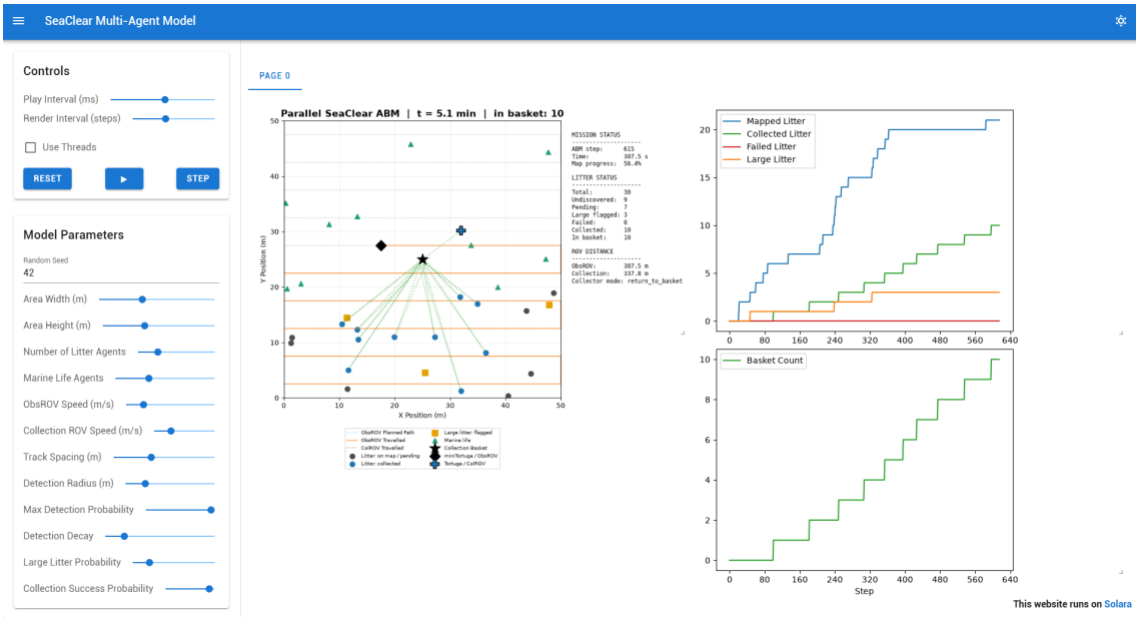


Figure 2.1 SeaClear Multi-Agent Model

Agent	Represents	What it does
ObservationROVAgent	miniTortuga	Follows a lawnmower search path and detects litter near it
CollectionROVAgent	Tortuga	Goes to the nearest mapped litter, picks it up and brings it to the basket
LitterAgent	Seabed litter	Does not move. Changes state when it is found or collected
BasketAgent	Collection basket	Stays in the middle of the area and counts the litter dropped in
MarineLifeAgent	Fish / sea life	Moves around randomly. Does not affect the robots yet

Observation ROV

The ObsROV moves along a lawnmower path, going back and forth across the area with 5 m between each track. Every step it moves 0.5 m and checks for litter within 3 m. Litter closer to the ROV is easier to detect. The chance of detection is $P(d) = P_{max} \times e^{-k \times d}$ where d is the distance to the litter, $P_{max} = 0.95$ is the highest chance, and $k = 0.18$ controls how fast the chance drops with distance. Once detected, the litter is added to the map.

Collection ROV

The ColROV starts at the basket. When it is free, it picks the nearest litter on the map, moves straight to it, and tries to pick it up. The pickup follows SeaClear's own target of a 90% collection success rate. If it works, the ColROV takes the litter back to the basket. If it fails, the litter is marked as failed, and the ColROV moves on to the next one.

Large litter

About 10% of the litter is randomly set as large. Large litter is still mapped but flagged, and the ColROV does not try to collect it because it is too big for the gripper.

Litter states

Each litter agent keeps track of its own state:

- UNDISCOVERED → ON_MAP (found by the ObsROV)
- ON_MAP → PICKED_UP → COLLECTED (picked up and dropped in the basket)
- ON_MAP → FAILED (pickup did not work)

Both ROVs move in the same model step, so mapping and collection happen at the same time. One step is 0.5 seconds of simulated time. The run stops when the ObsROV has finished its path and there is no more litter left for the ColROV to collect.

3. Results

Fig. 2.1 shows a run with the default settings (30 litter, 10 marine life, seed 42) at about 5 minutes in. The ObsROV has covered 56% of its path and mapped 21 litter, and the ColROV has already put 10 in the basket. This shows both robots working at the same time. We also ran the default settings 30 times with different seeds.

Result	Average
Litter mapped	30 out of 30
Litter collected into basket	24.2
Failed pickups	2.7
Large litter flagged	3.1
Time for ObsROV to finish searching	545 s (about 9 min)
Total mission time	825 s (about 14 min)

The ObsROV always finished searching first, and the ColROV needed about 4 to 5 more minutes to finish collecting. This means that with only one collector, collection is the slower part of the mission. We also tried changing the track spacing at 5 m and all litter was found, but at 8 m to 10 m only around 60% to 70% was found, because the gaps between tracks became bigger than the detection range.

4. References

Ilioudi, A., Sosnowski, S., Banken, E., Bevanda, P., Brüdigam, J., Buşoniu, L., Chardard, Y., Delea, C., De Schutter, B., Đuraš, A., Hertel-ten Eikelder, C., Heshmati-Alamdari, S., Maer, V.-M., Palunko, I., Pozniak, I., Prkačin, V., & Tolić, D. (2026). The SeaClear system: An intelligent multi-robot solution for autonomous cleanup of marine debris on the seabed. *Engineering Applications of Artificial Intelligence, 170*, Article 114094. https://doi.org/10.1016/j.engappai.2026.114094

Github Repository: https://github.com/syedmuhdhafidz/seaclear-mesa-abm